F78PA Probability & Statistics A

Tutorial 4

1. Consider the following game in which a player pays £1 to roll two fair 6-sided die.

   - If 2 sixes are obtained the player wins a cash prize of £10.
   - if 1 six is obtained the player wins £2.
   - Otherwise the player wins nothing.

   Let W represent the player's net profit from playing the game. Write down the probability mass function of the random variable W, and hence compute $\mathbb{E}(W)$.

   Suppose now that the player pays £c to play the game, where c is some constant, and that the prizes remain as before. Identify the value of c for which the player's expected net profit is zero.

2. A discrete random variable R has probability mass function $p_R(r)$ given by

r	1	2	3	4	5
$p_R(r)$	0.1	0.1	0.3	0.3	0.2

   Find the expected value and variance of the random variable $S = \log(R)$.

3. For $X \sim \text{Bin}(n, p)$, with $n \geq 1$ and $0 \leq p \leq 1$, show that $\mathbb{E}(X) = np$ and $\mathbb{V}\text{ar}(X) = np(1-p)$.

   [**Hint**: To prove this, you can first show that $\mathbb{E}(X(X-1)) = n(n-1)p^2$]

4. Suppose that X is a random variable with $\mathbb{E}[X] = \frac{3}{5}$ and with density function

$$f(x) = \begin{cases} a + bx & 0 < x < 1, \\ 0 & \text{otherwise,} \end{cases}$$

   and let $Y = X^2$.

   Determine

   (a) The values of a and b
   (b) $\mathbb{P}(1/4 \leq X \leq 1/2)$
   (c) $\mathbb{V}\text{ar}[X]$
   (d) $\mathbb{P}(1/4 \leq Y \leq 1/2)$
   (e) $\mathbb{V}\text{ar}[Y]$

5. A petrol station is supplied with petrol once a week. The weekly volume of sales in thousands of litres is a random variable X with probability density function

$$f(x) = \begin{cases} 5(1-x)^4 & 0 < x < 1, \\ 0 & \text{otherwise.} \end{cases}$$

   Determine how large the station's storage tank needs to be in order that the probability that the station runs out of petrol during the week is 0.01.

6. An archer shoots an arrow at a circular target of radius 1 metre in such a way that the probability that the arrow hits a region A on the target is proportional to the area of the region A (i.e. the arrow is equally likely to hit any point on the target). In addition, suppose that the archer never misses the target.

   (a) Determine the probability that the arrow lands closer to the centre of the circular target than to the edge of the target.

(b) Let X denote the distance, in centimetres, between where the arrow hits the target and the centre of the target. Determine $\mathrm{Im}(X)$, the image of X, and find the distribution function of X and sketch it.

[**Hint:** *To work out the distribution function of X, draw a picture of the target and sketch the region on the target which corresponds to the event $\{X \leq x\}$ and then figure out the probability of this event.*]

Determine $\mathbb{P}(30 < X < 90)$.

Determine $\mathbb{E}(X)$. Does this value seem reasonable? Why or why not?

7. Let X be a random variable with density

$$f_X(x) = \begin{cases} ax^2 & 0 \leq x \leq 4, \\ 0 & \text{otherwise.} \end{cases}$$

(a) Find a.

(b) Let $Y = 1/\sqrt{X}$. Determine $\mathrm{Im}(Y)$, the image of Y, and find f_Y, the density function of Y. (Note: You are not required to compute the distribution function F_Y of Y.)

8. Let X be a random variable with density

$$f_X(x) = \begin{cases} |x| & -1 \leq x \leq 1, \\ 0 & \text{otherwise.} \end{cases}$$

(a) Sketch the graph of f_X. Find $\mathbb{E}[X]$ and $\mathbb{V}\mathrm{ar}[X]$. Is your value for $E[X]$ intuitively reasonable?

(b) Let $Y = |X|$. Find the expectation and variance of Y.

Remark: This example shows that, in general, $\mathbb{E}(|X|) \neq |\mathbb{E}X|$.

9. Let X be a random variable with density

$$f_X(x) = \begin{cases} \frac{x^2}{9} & 0 \leq x \leq 3, \\ 0 & \text{otherwise.} \end{cases}$$

(a) Find $\mathbb{E}[X]$.

(b) Let $Y = 1/X$. Determine $\mathbb{E}[Y]$.

Remark: This example shows that, in general, $\mathbb{E}\left[\frac{1}{X}\right] \neq \frac{1}{\mathbb{E}[X]}$.

10. Suppose that X is an exponential random variable with parameter $\lambda > 0$.

(a) Show that $\mathbb{V}\mathrm{ar}[X] = 1/\lambda^2$

(b) Show that for any $c > 0$ the random variable cX is also an exponential random variable, with parameter λ/c.

Exam Style Question

(a) Consider the function

$$f_X(x) = \begin{cases} 2x - x^2, & 0 \leq x \leq 1, \\ \frac{5-3x}{2}, & 1 < x \leq \frac{5}{3}, \\ 0, & \text{otherwise.} \end{cases}$$

i) Sketch the graph of f_X and explain why it is a valid density function for a random variable X, carefully justifying your answer. [3 marks]

ii) Find $F_X(x)$, the distribution function of X. [3 marks]

iii) Determine $\mathbb{P}(X > \frac{3}{2} | X > 1)$. [2 marks]

iv) Find $\mathbb{E}[X]$, the expected value of X. [2 marks]

(b) Suppose that W follows an exponential distribution with parameter 2, that is, $W \sim \text{Exp}(2)$, and that $Y = 3W - 1$. Find the expected value, $\mathbb{E}[Y]$, and variance, $\mathbb{V}\text{ar}(Y)$, of Y. [2 marks]

[Total 12 marks]